

12.9 Events

Table 12-28 shows the events that are generated and the numbers that the PMU uses to reference the events. The table also shows the bit position of each event on the event bus. Event reference numbers that are not listed are reserved.

Table 12-28 PMU events

Event number	Event mnemonic	PMU event bus (to external)	PMU event bus (to trace)	Event name
0x00	SW_INCR	-	-	Software increment. The register is incremented only on writes to the Software Increment Register.
0x01	L1I_CACHE_REFILL	[0]	[0]	L1 Instruction cache refill.
0x02	L1I_TLB_REFILL	[1]	[1]	L1 Instruction TLB refill.
0x03	L1D_CACHE_REFILL	[2]	[2]	L1 Data cache refill.
0x04	L1D_CACHE	[3]	[3]	L1 Data cache access.
0x05	L1D_TLB_REFILL	[4]	[4]	L1 Data TLB refill.
0x06	LD_RETIRED	[5]	[5]	Instruction architecturally executed, condition check pass - load.
0x07	ST_RETIRED	[6]	[6]	Instruction architecturally executed, condition check pass - store.
0x08	INST_RETIRED	[7]	[7]	Instruction architecturally executed.
-	-	[8]	[8]	Two instructions architecturally executed. Counts every cycle in which two instructions are architecturally retired. Event 0x08, INST_RETIRED, always counts when this event counts.
0x09	EXC_TAKEN	[9]	[9]	Exception taken.
0x0A	EXC_RETURN	[10]	[10]	Exception return.
0x0B	CID_WRITE_RETIRED	[11]	[11]	Change to Context ID retired.
0x0C	PC_WRITE_RETIRED	[12]	[12]	Instruction architecturally executed, condition check pass, software change of the PC.
0x0D	BR_IMMED_RETIRED	[13]	[13]	Instruction architecturally executed, immediate branch.
0x0E	BR_RETURN_RETIRED	-	-	Instruction architecturally executed, condition code check pass, procedure return.
0x0F	UNALIGNED_LDST_RETIRED	[14]	[14]	Instruction architecturally executed, condition check pass, unaligned load or store.
0x10	BR_MIS_PRED	[15]	[15]	Mispredicted or not predicted branch speculatively executed.

Table 12-28 PMU events (continued)

Event number	Event mnemonic	PMU event bus (to external)	PMU event bus (to trace)	Event name
0x11	CPU_CYCLES	-	-	Cycle.
0x12	BR_PRED	[16]	[16]	Predictable branch speculatively executed.
0x13	MEM_ACCESS	[17]	[17]	Data memory access.
0x14	L1I_CACHE	[18]	[18]	L1 Instruction cache access.
0x15	L1D_CACHE_WB	[19]	[19]	L1 Data cache Write-Back.
0x16	L2D_CACHE	[20]	[20]	L2 Data cache access.
0x17	L2D_CACHE_REFILL	[21]	[21]	L2 Data cache refill.
0x18	L2D_CACHE_WB	[22]	[22]	L2 Data cache Write-Back.
0x19	BUS_ACCESS	-	-	Bus access.
0x1A	MEMORY_ERROR	-	-	Local memory error.
0x1D	BUS_CYCLES	-	-	Bus cycle.
0x1E	CHAIN	-	-	Odd performance counter chain mode.
0x60	BUS_ACCESS_LD	-	-	Bus access - Read.
0x61	BUS_ACCESS_ST	-	-	Bus access - Write.
0x7A	BR_INDIRECT_SPEC	-	-	Branch speculatively executed - Indirect branch.
0x86	EXC_IRQ	-	-	Exception taken, IRQ.
0x87	EXC_FIQ	-	-	Exception taken, FIQ.
0xC0	-	-	-	External memory request.
0xC1	-	-	-	Non-cacheable external memory request.
0xC2	-	-	-	Linefill because of prefetch.
0xC3	-	-	-	Instruction Cache Throttle occurred.
0xC4	-	-	-	Entering read allocate mode.
0xC5	-	-	-	Read allocate mode.
0xC6	-	-	-	Pre-decode error.
0xC7	-	-	-	Data Write operation that stalls the pipeline because the store buffer is full.
0xC8	-	-	-	SCU Snooped data from another CPU for this CPU.
0xC9	-	-	-	Conditional branch executed.
0xCA	-	-	-	Indirect branch mispredicted.

Table 12-28 PMU events (continued)

Event number	Event mnemonic	PMU event bus (to external)	PMU event bus (to trace)	Event name
0xCB	-	-	-	Indirect branch mispredicted because of address miscompare.
0xCC	-	-	-	Conditional branch mispredicted.
0xD0	-	[23]	[23]	L1 Instruction Cache (data or tag) memory error.
0xD1	-	[24]	[24]	L1 Data Cache (data, tag or dirty) memory error, correctable or non-correctable.
0xD2	-	[25]	[25]	TLB memory error.
0xE0	-	-	-	Attributable Performance Impact Event. Counts every cycle that the DPU IQ is empty and that is not because of a recent micro-TLB miss, instruction cache miss or pre-decode error.
0xE1	-	-	-	Attributable Performance Impact Event. Counts every cycle the DPU IQ is empty and there is an instruction cache miss being processed.
0xE2	-	-	-	Attributable Performance Impact Event. Counts every cycle the DPU IQ is empty and there is an instruction micro-TLB miss being processed.
0xE3	-	-	-	Attributable Performance Impact Event. Counts every cycle the DPU IQ is empty and there is a pre-decode error being processed.
0xE4	-	-	-	Attributable Performance Impact Event. Counts every cycle there is an interlock that is not because of an Advanced SIMD or Floating-point instruction, and not because of a load/store instruction waiting for data to calculate the address in the AGU. Stall cycles because of a stall in Wr, typically awaiting load data, are excluded.
0xE5	-	-	-	Attributable Performance Impact Event. Counts every cycle there is an interlock that is because of a load/store instruction waiting for data to calculate the address in the AGU. Stall cycles because of a stall in Wr, typically awaiting load data, are excluded.

Table 12-28 PMU events (continued)

Event number	Event mnemonic	PMU event bus (to external)	PMU event bus (to trace)	Event name
0xE6	-	-	-	Attributable Performance Impact Event. Counts every cycle there is an interlock that is because of an Advanced SIMD or Floating-point instruction. Stall cycles because of a stall in the Wr stage, typically awaiting load data, are excluded.
0xE7	-	-	-	Attributable Performance Impact Event Counts every cycle there is a stall in the Wr stage because of a load miss.
0xE8	-	-	-	Attributable Performance Impact Event. Counts every cycle there is a stall in the Wr stage because of a store.
-	-	[26]	-	L2 (data or tag) memory error, correctable or non-correctable.
-	-	[27]	-	SCU snoop filter memory error, correctable or non-correctable.
-	-	[28]	-	Advanced SIMD and Floating-point retention active.
-	-	-	-	CPU retention active.